

Midsem - I

P1.

$$\delta H = K \delta(x - a/2) \quad ; \quad K = \frac{\hbar^2 k^2}{4ma}$$

$$\psi_n^0(x) = \langle x | n^0 \rangle = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right) \quad ; \quad n=1, 2, \dots$$

$$E_n^0 = \frac{n^2 \hbar^2 k^2}{2ma^2}$$

Q (1) :

$$E_n^1 = \langle n | \delta H | n \rangle$$

$$= \int_0^a dx \quad \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right) K \delta(x - a/2)$$

1

E_n^1

$$= \frac{2K}{a} \sin^2\left(\frac{n\pi}{2}\right) = \begin{cases} \frac{2K}{a} & ; n, \text{ odd} \\ 0 & ; n, \text{ even} \end{cases}$$

$$\langle x | n' \rangle = - \sum_{m \neq n} \frac{\langle m | \delta H | n \rangle}{E_m^0 - E_n^0} \langle x | m \rangle$$

$$\begin{aligned} * \quad \langle m | \delta H | n \rangle &= \int_0^a dx \frac{2}{a} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi x}{a}\right) \\ &\quad \times \delta(x - a/2) \end{aligned}$$

$$= \frac{2k}{a} \sin\left(\frac{m\pi}{2}\right) \sin\left(\frac{n\pi}{2}\right)$$

$$* \quad E_m^0 - E_n^0 = (m^2 - n^2) \frac{\hbar^2 \pi^2}{2ma^2}$$

$$\langle x | n' \rangle = - \sum_{m \neq n} \frac{\sin\left(\frac{m\pi}{2}\right) \sin\left(\frac{n\pi}{2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi x}{a}\right)}{m^2 - n^2}$$

$$\Rightarrow \langle x | n' \rangle = \begin{cases} - \sum_{m \neq n} \frac{(+/-)}{m^2 - n^2} \sin\left(\frac{m\pi x}{2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi x}{a}\right) : n \text{ odd} \\ 0 : n \text{ even} \end{cases} \quad \boxed{2}$$

$\Rightarrow$ to 0 (1), all "n even states" remain unperturbed.

Reason: at $x = a/2$ where $\psi_H \neq 0$,

$$\langle x | n \rangle = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{2}\right) = 0 \quad \text{for } n \text{ even}$$

$\boxed{1}$

Ground state at $0(\lambda)$:

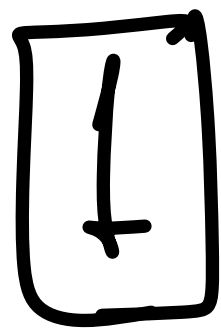
$$\langle x|1\rangle = \langle x|1^0\rangle + \langle x|1^1\rangle$$

$$= \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a} - \sum_{m \neq 1} \frac{\sin \left(\frac{m\pi}{2} \right)}{(m^2 - 1)} \langle x|m^0\rangle$$

$$= \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a} - \sum_{m=3,5,\dots} \frac{(+/-)}{m^2 - 1} \langle x|m^0$$

$$= \sqrt{\frac{2}{a}} \left\{ \sin \frac{\pi x}{a} + \frac{1}{8} \sin \frac{3\pi x}{a} - \frac{1}{24} \sin \frac{5\pi x}{a} \right.$$

$$\left. + \dots \right\} //$$



P2.

$$E_{f1} = E_n^0 \left(1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right)$$

$$\text{for } n=3, \quad l=0, 1, 2 \Rightarrow j = |l-1|, \dots, (l+1)$$

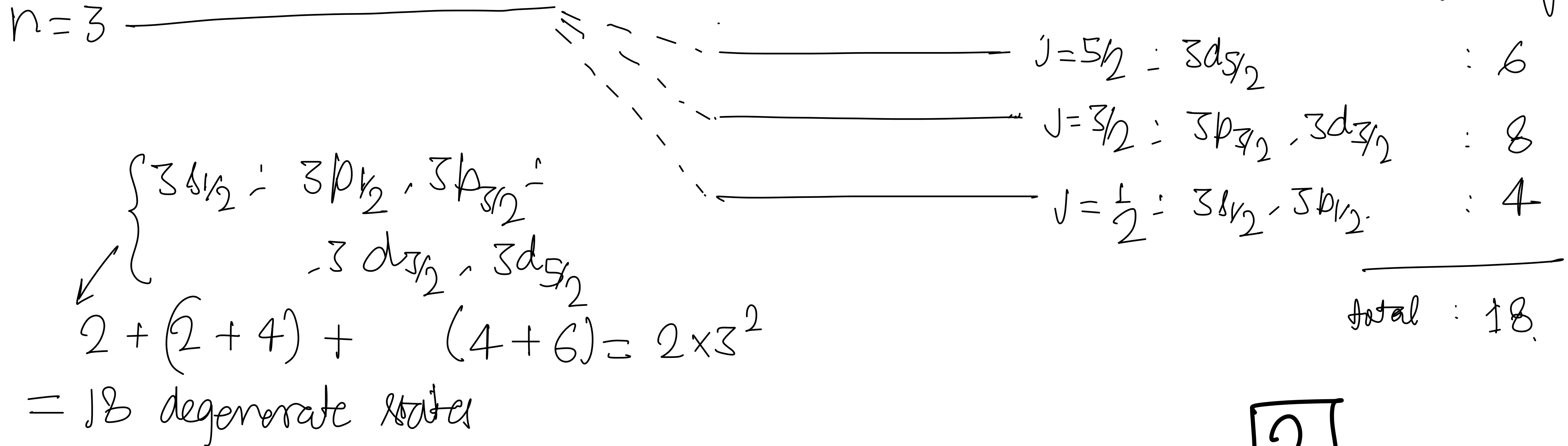
$$= \frac{1}{2}, \frac{3}{2}, \frac{5}{2} \quad \boxed{1}$$

Since E_{f1} is j -dependent, E_{f1}
little degeneracy in j but not in m_j . $\boxed{1}$

$$\text{Also, } E_{3,1/2} > E_{3,3/2} > E_{3,5/2}. \quad \boxed{1}$$

$$\delta H_{fs} = 0$$

$$\delta H_{fs} \neq 0.$$



2

P3

a)

True.

The difference arises due to different reduced mass of hydrogen & deuterium.

$$\therefore \Delta E_{f1} \propto \mu \quad \text{and} \quad \mu_D = \frac{m_e}{1 + m_e/2m_p} = \left(\frac{m_e}{1 + \frac{m_e}{m_p}} \right) \cdot \frac{1 + m_e/m_p}{(1 + m_e/2m_p)}$$

2

$$= \mu_H \left(\frac{1 + m_e/m_p}{1 + m_e/2m_p} \right) > \mu_H$$

b)

False.

$$S_{Hnf} = 0$$

$$\text{degeneracy} = 2 \times 2 = 4$$

for e^- spin for p spin

$$S_{Hnf} \neq 0. \quad \nearrow \text{singlet}$$

$$2 \otimes 2 = 0 \oplus 1 \quad \searrow \text{triplet}$$

$S=1 (3)$

$S=0 (1)$

2

i.e. $S=1$ state still has 3 fold degeneracy.

c) True:

The result of first order perturbation theory is valid provided we work with diagonal basis. In that basis,

$$|n^{\pm}\rangle = \sum_{m \notin D} \frac{\langle m | \delta H | n \rangle}{E_m^0 - E_n^0} |m\rangle$$

dimension of degenerate space.

2

$\Rightarrow$ if dimension of degenerate subspace $\geq$ that of Hilbert space,
no state is available on the R.H.S.

$$\Rightarrow |n^{\pm}\rangle = 0.$$

d) Falke.

$$H^0 = p^2/2m :$$

$$\psi(x) = e^{\pm i p \cdot x}$$

↓
continuous spectrum

$$\delta H = \frac{1}{2} m \omega^2 x^2$$

↓
discrete spectrum

[2]

Application of perturbation theory assumes that,

$$\lim_{\lambda \rightarrow 0} |n^1\rangle = |n^0\rangle$$

This does not hold for $H = H^0 + \lambda \delta H$

c) False.

$$\delta H \propto \vec{L} \cdot \vec{S} \quad : H^0 = \frac{p^2}{2m} - \frac{e^2}{r} \\ = \frac{1}{2} (\sigma^2 - L^2 - S^2)$$

clearly, $[\vec{S}, H^0] = [\vec{S}, \delta H] = 0$

$\Rightarrow$ rotational symmetry is preserved.

however, $[\vec{L}, \delta H] \neq 0$

$\& [\vec{S}, \delta H] \neq 0.$

2